

Midterm Activity #5: Personal Expense Tracker

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Course, Year & Section: BSIT 2-3

Course Code & Description: COMP 009 Object Oriented Programming

Submitted to: Sir SEVERINO M. BEDIS JR.

General Instructions

1. Create a Java program named `ExpenseTrackerLastName`.
2. Implement the following methods:
 - A void method that displays the program title.
 - A non-void method that calculates the total expenses using three values:
 - `food`
 - `transportation`
 - `other expenses`
 - A non-void method that checks if the total expense exceeds a budget and returns a message.
 - A void method that displays the total expense and the budget status.
3. In the main method:
 - Call all the methods you created.
 - Display the final output clearly.
4. Use appropriate method names, parameters, and return types.

5. Add at least TWO enhancement:

- Accept user input
- Add another expense category
- Compute remaining budget
- Display a personalized message
- 6. Ensure your program:
 - Compiles without errors
 - Produces correct output

Submission Format

- Submit the Java source code file (.java).
- The filename must follow this format: ExpenseTrackerLastName.pdf or ExpenseTrackerLastName.docx
- Include screenshots of the program output showing the following:
 - User input
 - Program processing
 - Final output/result
- The screenshot must clearly display the entire console output.

Academic Integrity Policy

- Students must write their own program.
- The following are not allowed:
 - Copying code from classmates.
 - Sharing source code.
 - Downloading solutions from the internet.
- Students with identical programs may receive a score of 0.

PART 1 of the screenshot User Input

```

11
12- public static String checkBdgtStatus(String name, double totalEXPN, double
    bdgt) {
13-     if (totalEXPN > bdgt) {
14-         return "WARNING!, " + name + "! HEY! Yes you! You have exceeded your
            budget. Time to cut back!";
15-     } else if (totalEXPN == bdgt) {
16-         return "OOPS! Careful Goat, " + name + ". You reached your budget
            limit. Utang next time!";
17-     } else {
18-         return "Great job u Da Man!, " + name + "! You managed your finances
            well and stayed under budget.";
19-     }
20- }
21
22- public static void displayFinancialSummary(double totalEXPN, double bdgt,
    double remainingBalance, String statmesg) {
23-     System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
24-     System.out.printf(" Total bdgt: $%.2f\n", bdgt);
25-     System.out.printf(" Total Expenses: $%.2f\n", totalEXPN);
26-     System.out.printf(" Remaining Balance: $%.2f\n", remainingBalance);
27-     System.out.println("-----");
28-     System.out.println(" Status: " + statmesg);
29-     System.out.println("=====");
30- }
31- public static void main(String[] args) {
32-     Scanner scanner = new Scanner(System.in);
33-     String choice;
34-
35-     do {
36-         displayProgramTitle();
37-         System.out.print("Please enter your name: ");
38-         String name = scanner.nextLine();
39-         double bdgt = getValidMonetaryInput(scanner, "Oke! Enter your total
            budget: $");

```

Output

```

!!-----PERSONAL EXPENSE TRACKER ni JAY-----!!
Please enter your name: Jay
Oke! Enter your total budget: $100
Enter your food expenses: $10
Enter transportation expenses: $10
Enter other expenses : $10
Enter your addition expenses ehem scatter: $1000

```

PART 2 of the screenshot Progress

```

11
12- public static String checkBdgtStatus(String name, double totalEXPN, double
    bdgt) {
13-     if (totalEXPN > bdgt) {
14-         return "WARNING!, " + name + "! HEY! Yes you! You have exceeded your
            budget. Time to cut back!";
15-     } else if (totalEXPN == bdgt) {
16-         return "OOPS! Careful Goat, " + name + ". You reached your budget
            limit. Utang next time!";
17-     } else {
18-         return "Great job u Da Man!, " + name + "! You managed your finances
            well and stayed under budget.";
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20- }
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22- public static void displayFinancialSummary(double totalEXPN, double bdgt,
    double remainingBalance, String statmesg) {
23-     System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
24-     System.out.printf(" Total bdgt: $%.2f\n", bdgt);
25-     System.out.printf(" Total Expenses: $%.2f\n", totalEXPN);
26-     System.out.printf(" Remaining Balance: $%.2f\n", remainingBalance);
27-     System.out.println("-----");
28-     System.out.println(" Status: " + statmesg);
29-     System.out.println("=====");
30- }
31- public static void main(String[] args) {
32-     Scanner scanner = new Scanner(System.in);
33-     String choice;
34-
35-     do {
36-         displayProgramTitle();
37-         System.out.print("Please enter your name: ");
38-         String name = scanner.nextLine();
39-         double bdgt = getValidMonetaryInput(scanner, "Oke! Enter your total
            budget: $");

```

Output

```

!!-----PERSONAL EXPENSE TRACKER ni JAY-----!!
Please enter your name: Jay
Oke! Enter your total budget: $100
Enter your food expenses: $10
Enter transportation expenses: $10
Enter other expenses : $10
Enter your addition expenses ehem scatter: $1000

!!-----FINANCIAL SUMMARY MO PO!-----!!

Total bdgt: $100.00
Total Expenses: $1030.00
Remaining Balance: $-930.00

-----
Status: WARNING!, Jay! HEY! Yes you! You have exceeded your budget. Time to cut back!
=====
Would you like to try again and track another person's expenses? (yes/no/n/exit):

```

PART 3 of the screenshot Results

```

11
12- public static String checkBdgtStatus(String name, double totalEXPN, double
    bdgt) {
13-     if (totalEXPN > bdgt) {
14-         return "WARNING!, " + name + "! HEY! Yes you! You have exceeded your
            budget. Time to cut back!";
15-     } else if (totalEXPN == bdgt) {
16-         return "OOPS! Careful Goat, " + name + ". You reached your budget
            limit. Utang next time!";
17-     } else {
18-         return "Great job u Da Man!, " + name + "! You managed your finances
            well and stayed under budget.";
19-     }
20- }
21
22- public static void displayFinancialSummary(double totalEXPN, double bdgt,
    double remainingBalance, String statmesg) {
23-     System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
24-     System.out.printf(" Total bdgt: $%.2f\n", bdgt);
25-     System.out.printf(" Total Expenses: $%.2f\n", totalEXPN);
26-     System.out.printf(" Remaining Balance: $%.2f\n", remainingBalance);
27-     System.out.println("-----");
28-     System.out.println(" Status: " + statmesg);
29-     System.out.println("=====");
30- }
31- public static void main(String[] args) {
32-     Scanner scanner = new Scanner(System.in);
33-     String choice;
34-
35-     do {
36-         displayProgramTitle();
37-         System.out.print("Please enter your name: ");
38-         String name = scanner.nextLine();
39-         double bdgt = getValidMonetaryInput(scanner, "Oke! Enter your total
            budget: $");

```

Output

```

!!-----PERSONAL EXPENSE TRACKER ni JAY-----!!
Please enter your names: Jay
Oke! Enter your total budget: $1000
Enter your food expenses: $10
Enter transportation expenses: $10
Enter other expenses : $10
Enter your addition expenses ehem scatter: $1000

!!-----FINANCIAL SUMMARY MO PO!-----!!

Total bdgt: $1000.00
Total Expenses: $1030.00
Remaining Balance: $-30.00

-----
Status: WARNING!, Jay! HEY! Yes you! You have exceeded your budget. Time to cut back!
=====
Would you like to try again and track another person's expenses? (yes/no/n/exit): yes
!!-----PERSONAL EXPENSE TRACKER ni JAY-----!!
Please enter your name: Jay 2
Oke! Enter your total budget: $1000
Enter your food expenses: $10
Enter transportation expenses: $10
Enter other expenses : $10
Enter your addition expenses ehem scatter: $100

!!-----FINANCIAL SUMMARY MO PO!-----!!

Total bdgt: $1000.00
Total Expenses: $130.00
Remaining Balance: $870.00

-----
Status: Great job u Da Man!, Jay 2! You managed your finances well and stayed under
budget.
=====

```

PART 4 of the screenshot Results

```

11
12- public static String checkBdgtStatus(String name, double totalEXPN, double
    bdgt) {
13-     if (totalEXPN > bdgt) {
14-         return "WARNING!, " + name + "! HEY! Yes you! You have exceeded your
            budget. Time to cut back!";
15-     } else if (totalEXPN == bdgt) {
16-         return "OOPS! Careful Goat, " + name + ". You reached your budget
            limit. Utang next time!";
17-     } else {
18-         return "Great job u Da Man!, " + name + "! You managed your finances
            well and stayed under budget.";
19-     }
20- }
21
22- public static void displayFinancialSummary(double totalEXPN, double bdgt,
    double remainingBalance, String statmesg) {
23-     System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
24-     System.out.printf(" Total bdgt: $%.2f\n", bdgt);
25-     System.out.printf(" Total Expenses: $%.2f\n", totalEXPN);
26-     System.out.printf(" Remaining Balance: $%.2f\n", remainingBalance);
27-     System.out.println("-----");
28-     System.out.println(" Status: " + statmesg);
29-     System.out.println("=====");
30- }
31- public static void main(String[] args) {
32-     Scanner scanner = new Scanner(System.in);
33-     String choice;
34-
35-     do {
36-         displayProgramTitle();
37-         System.out.print("Please enter your name: ");
38-         String name = scanner.nextLine();
39-         double bdgt = getValidMonetaryInput(scanner, "Oke! Enter your total
            budget: $");

```

Output

```

Enter other expenses : $10
Enter your addition expenses ehem scatter: $100

!!-----FINANCIAL SUMMARY MO PO!-----!!

Total bdgt: $1000.00
Total Expenses: $130.00
Remaining Balance: $870.00

-----
Status: Great job u Da Man!, Jay 2! You managed your finances well and stayed under
budget.
=====
Would you like to try again and track another person's expenses? (yes/no/n/exit): yes
!!-----PERSONAL EXPENSE TRACKER ni JAY-----!!
Please enter your name: Jay 3
Oke! Enter your total budget: $1000
Enter your food expenses: $100
Enter transportation expenses: $100
Enter other expenses : $100
Enter your addition expenses ehem scatter: $700

!!-----FINANCIAL SUMMARY MO PO!-----!!

Total bdgt: $1000.00
Total Expenses: $1000.00
Remaining Balance: $0.00

-----
Status: OOPS! Careful Goat, Jay 3. You reached your budget limit. Utang next time!
=====
Would you like to try again and track another person's expenses? (yes/no/n/exit): no
Thank you for using my very own Personal Expense Tracker. Goodbye GOAT!!!

```

Code

```

import java.util.Scanner;

public class ExpenseTrackerArevalo {

    public static void displayProgramTitle() {
        System.out.println("!!-------PERSONAL EXPENSE TRACKER ni JAY-----!!");
    }

    public static double calculatetotalEXPNs(double food, double transport, double othr, double ent) {
        return food + transport + othr + ent;
    }

    public static String checkbdgtStatus(String name, double totalEXPN, double bdgt) {
        if (totalEXPN > bdgt) {
            return "WARNING!, " + name + "! HEY! Yes you! You have exceeded your budget. Time to cut back!";
        } else if (totalEXPN == bdgt) {
            return "OOPS! Careful Goat, " + name + ". You reached your budget limit. Utang next time!";
        } else {
            return "Great job u Da Man!, " + name + "! You managed your finances well and stayed under budget.";
        }
    }

    public static void displayFinancialSummary(double totalEXPN, double bdgt, double remainblnc, String statmesg) {
        System.out.println("\n!-------FINANCIAL SUMMARY MO PO!-------!!\n");
        System.out.printf(" Total bdgt: $%.2f\n", bdgt);
        System.out.printf(" Total Expenses: $%.2f\n", totalEXPN);
        System.out.printf(" Remaining Balance: $%.2f\n", remainblnc);
        System.out.println("-----");
        System.out.println(" Status: " + statmesg);
        System.out.println("=====");
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String choice;

        do {
            displayProgramTitle();
            System.out.print("Please enter your name: ");
            String name = scanner.nextLine();
            double bdgt = getValidMonetaryInput(scanner, "Oke! Enter your total budget: $");
            double food = getValidMonetaryInput(scanner, "Enter your food expenses: $");
            double transport = getValidMonetaryInput(scanner, "Enter transportation expenses: $");
            double othr = getValidMonetaryInput(scanner, "Enter other expenses : $");
            double ent = getValidMonetaryInput(scanner, "Enter your addiction expenses ehem scatter: $");
            double totalEXPNs = calculatetotalEXPNs(food, transport, othr, ent);
            double remainblnc = bdgt - totalEXPNs;
            String message = checkbdgtStatus(name, totalEXPNs, bdgt);
            displayFinancialSummary(totalEXPNs, bdgt, remainblnc, message);
            System.out.print("Would you like to try again and track another person's expenses? (yes/no/n/exit): ");
            choice = scanner.nextLine().trim().toLowerCase();

        } while (!(choice.equals("no") || choice.equals("n") || choice.equals("exit")));
    }
}

```

```
        System.out.println("Thank you for using my very own Personal Expense Tracker. Goodbye GOAT!!!");
        scanner.close();
    }
    private static double getValidMonetaryInput(Scanner scanner, String prompt){
        double amount;
        while(true){
            System.out.print(prompt);
            if (scanner.hasNextDouble()){
                amount = scanner.nextDouble();
                scanner.nextLine();

                if (amount >= 0){
                    break;
                } else {
                    System.out.println("--> Invalid man: Amount cannot be negative. IT CANNOT BE AN UTANG! Try again.");
                }
            } else {
                System.out.println("--> Invalid: Please enter an actual numeric value.");
                scanner.next();
                scanner.nextLine();
            }
        }
        return amount;
    }
}
```